Full-Stack & AI/ML Developer • B.Tech CSE (AI-Driven DevOps)

Bengaluru, Karnataka | +91 77506 08944 | ghoul.raider13@gmail.com
github.com/ghoulraider13-rgb | linkedin.com/in/ashwin-lahkar-08a9ab405

SUMMARY

Full-stack engineer with strong AI/ML background. Deploy lightweight vision models on edge devices, build privacy-preserving browser agents, and create AI-powered productivity tools. Proficient in React, Python, TypeScript, and modern ML toolchains (YOLO, ONNX, TensorRT, Genetic Algorithms). Open-source contributor (npm package **actis**, Hermes Agent skills). Smart India Hackathon 2026 participant (ISRO PS 26171 — Team Sahayak).

EDUCATION

Jain University — B.Tech Computer Science (AI-Driven DevOps)

2024–2028 • CGPA: 9.2/10 (expected) • Bengaluru, Karnataka

TECHNICAL SKILLS

Languages	Python, JavaScript/TypeScript, Java, C, SQL
AI/ML & CV	YOLOv11/ONNX/TensorRT/OpenVINO, Genetic Algorithms, LLM Integration (Gemini, Nemotron), NLP,
Web & Frontend	React 18, Vite, Tailwind CSS, Framer Motion, Chrome MV3 Extensions, PWA, Capacitor, Tauri
Backend & APIs	FastAPI, Node.js/Express, REST APIs, WebSockets, Server-Sent Events
DevOps & Tools	Git/GitHub, GitHub Actions, Docker, Linux CLI, VS Code, Jupyter, Roboflow, Google Cloud
Databases	SQLite (sql.js), localStorage, IndexedDB, PostgreSQL (basics)

SELECTED PROJECTS

React • Vite • Tailwind • Framer Motion • Gemini API • Nemotron • Tauri • Capacitor • PWA

Full-featured task manager with natural language parsing (Gemini/NVIDIA Nemotron), Pomodoro timer, enforcer-mode alarms, rich scratchpad with draw tool, sleep & hydration trackers. Live at flotask-xi.vercel.app with native desktop (Tauri) and mobile (Capacitor) builds. Responsive single-page layout with reactive magnetic background.

Tech: React 18, Vite, Tailwind, Framer Motion, Gemini Flash, NVIDIA Nemotron, Tauri, Capacitor, PWA, localStorage

Links: https://flotask-xi.vercel.app | github.com/ghoulraider13-rgb/FloTask

Node.js • TypeScript • ESM • sql.js • chokidar • SQLite

CLI-first tool with local SQLite ledger, debounced filesystem watcher, natural-language query parser, and plain-Node dashboard (port 7420). Zero external dependencies, pure ESM. Published to npm as **actis**. Features rule-based NL query parser and real-time web UI.

Tech: Node.js, TypeScript, ESM, sql.js, chokidar, SQLite, npm

Links: npm: actis | github.com/ghoulraider13-rgb/Actis | localhost:7420

YOLOv11 • ONNX • TensorRT • OpenVINO • Python • Roboflow • Jetson

YOLOv11n models for flood, landslide, and earthquake damage detection. ONNX, TensorRT, and OpenVINO optimization for Jetson-class UAV edge deployment. IEEE-style evaluation with mAP/latency benchmarks across optimization backends. Research paper in preparation.

Tech: YOLOv11, ONNX Runtime, TensorRT, OpenVINO, Python, Roboflow, NVIDIA Jetson

Links: github.com/ghoulraider13-rgb/Disaster_DetectionDrone

Chrome MV3 • FastAPI • Python • PII Redaction • Privacy Engineering

Chrome Manifest V3 extension + FastAPI backend for privacy-preserving web task automation. PII redaction engine achieving 1.00 recall / 1.00 precision on adversarial tests. ISRO Problem Statement 26171 — internal round build. End-to-end encrypted local processing.

Tech: Chrome MV3, FastAPI, Python, PII Redaction, Encryption, Privacy

Links: Private repo (ISRO PS 26171)

Python • Genetic Algorithms • scikit-learn • RandomForest • Optimization

Genetic algorithm reduces feature set by 50% while raising RandomForest accuracy from 97.1% to 99.4% on breast-cancer dataset. Demonstrates evolutionary computation for automated feature engineering with tournament selection, crossover, and adaptive mutation.

Tech: Python, DEAP/GA, scikit-learn, RandomForest, NumPy, Pandas

Links: Private repo

TypeScript • Node.js • MCP • CDP • AI Agents • Automation

Authored and maintain 20+ skills for Hermes Agent (Nous Research): autonomous coding agents (Claude Code, Codex, OpenCode, Computer Use), desktop automation, web scraping/recovery, research tools, media processing, and dev tooling. Used in production agent workflows across multiple projects.

Tech: TypeScript, Node.js, MCP, Chrome DevTools Protocol, AI Agent Orchestration

Links: hermes-agent.nousresearch.com/docs

CERTIFICATIONS & ACHIEVEMENTS

- Smart India Hackathon 2026 — ISRO PS 26171 (Team Sahayak) — Internal round build & PII redaction engine
- Google Cloud Study Jam — Core GCP labs (Compute, Storage, Networking, IAM, Monitoring)
- Coursera — Operating Systems (3 courses), Data Structures in C, Python Crash Course
- Published npm package: **actis** (filesystem activity ledger, CLI + web dashboard)
- Hermes Agent contributor — 20+ skills for autonomous AI agent workflows (Nous Research)